OpenCL for FPGA: A HLS Framework

Thursday, April 9, 2026 4:10 PM

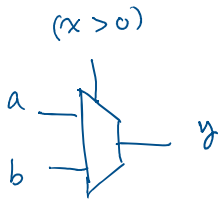
- Data Path. from sequential code

  mem[A] += mem[B] * C ⇒ sum += a[i] * b;

- Synthesize if statements

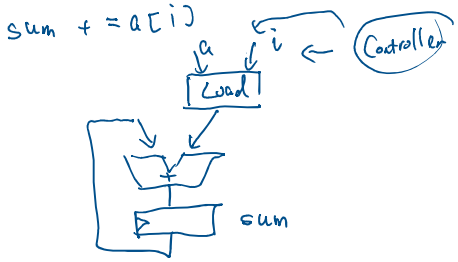
```
if ( x > 0)
    y = a;
else
    y = b;
```

⇒



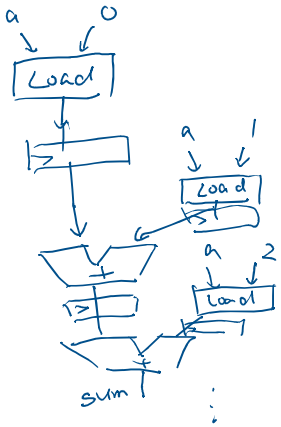
- While / for loop.
  - controller → FSM / Counter.
  - unroll

eg. for (i=0; i<N; i++)
    sum += a[i];



unrolling: higher parallelism (+) (MLP, PLP)
higher area (→)

sum += a[0];
sum += a[1];
sum += a[2];
⇒



Synthesize from C / Java / C++

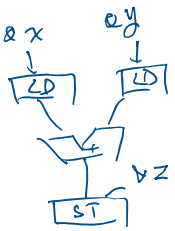
↓

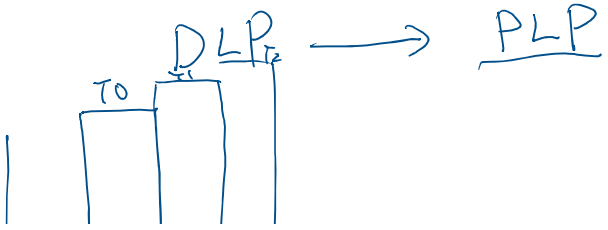
Synthesize from CUDA / OpenCL → DLP represented as TLP

Explicit DLP / TLP

eg. C[tid] = A[tid] + B[tid];

// Z = X + y. ⇒






time